

```
In [1]: import pandas as pd  
import numpy as np
```

```
In [2]: df = pd.read_csv(r"C:\Users\LIKHITH\OneDrive\Desktop\Data science\Data_files\Salary  
df
```

Out[2]:

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891
5	2.9	56642
6	3.0	60150
7	3.2	54445
8	3.2	64445
9	3.7	57189
10	3.9	63218
11	4.0	55794
12	4.0	56957
13	4.1	57081
14	4.5	61111
15	4.9	67938
16	5.1	66029
17	5.3	83088
18	5.9	81363
19	6.0	93940
20	6.8	91738
21	7.1	98273
22	7.9	101302
23	8.2	113812
24	8.7	109431
25	9.0	105582
26	9.5	116969
27	9.6	112635
28	10.3	122391
29	10.5	121872

Mean

```
In [3]: df.mean()
```

```
Out[3]: YearsExperience    5.313333  
Salary                76003.000000  
dtype: float64
```

```
In [6]: df['Salary'].mean()
```

```
Out[6]: np.float64(76003.0)
```

Median

```
In [7]: df.median()
```

```
Out[7]: YearsExperience    4.7  
Salary                65237.0  
dtype: float64
```

```
In [8]: df['Salary'].median()
```

```
Out[8]: 65237.0
```

Mode

```
In [9]: df.mode()
```

Out[9]:

	YearsExperience	Salary
0	3.2	37731
1	4.0	39343
2	NaN	39891
3	NaN	43525
4	NaN	46205
5	NaN	54445
6	NaN	55794
7	NaN	56642
8	NaN	56957
9	NaN	57081
10	NaN	57189
11	NaN	60150
12	NaN	61111
13	NaN	63218
14	NaN	64445
15	NaN	66029
16	NaN	67938
17	NaN	81363
18	NaN	83088
19	NaN	91738
20	NaN	93940
21	NaN	98273
22	NaN	101302
23	NaN	105582
24	NaN	109431
25	NaN	112635
26	NaN	113812
27	NaN	116969
28	NaN	121872
29	NaN	122391

```
In [10]: df['Salary'].mode()
```

```
Out[10]: 0      37731
         1      39343
         2      39891
         3      43525
         4      46205
         5      54445
         6      55794
         7      56642
         8      56957
         9      57081
        10      57189
        11      60150
        12      61111
        13      63218
        14      64445
        15      66029
        16      67938
        17      81363
        18      83088
        19      91738
        20      93940
        21      98273
        22     101302
        23     105582
        24     109431
        25     112635
        26     113812
        27     116969
        28     121872
        29     122391
        Name: Salary, dtype: int64
```

Variance

```
In [11]: df.var()
```

```
Out[11]: YearsExperience    8.053609e+00
         Salary             7.515510e+08
         dtype: float64
```

```
In [12]: df['Salary'].var()
```

```
Out[12]: 751550960.4137931
```

Standard deviation

```
In [13]: df.std()
```

```
Out[13]: YearsExperience    2.837888
Salary    27414.429785
dtype: float64
```

```
In [14]: df['Salary'].std()
```

```
Out[14]: 27414.4297845823
```

Coefficient of variance(cv)

```
In [15]: # for calculating cv we have to import a library first
from scipy.stats import variation
variation(df.values) # this will give cv of entire dataframe
```

```
Out[15]: array([0.5251297 , 0.35463929])
```

```
In [16]: variation(df['Salary']) # this will give us cv of that particular column
```

```
Out[16]: np.float64(0.3546392938275572)
```

Correlation

```
In [17]: df.corr()
```

```
Out[17]:
```

	YearsExperience	Salary
YearsExperience	1.000000	0.978242
Salary	0.978242	1.000000

```
In [19]: df['Salary'].corr(df['YearsExperience'])
```

```
Out[19]: np.float64(0.9782416184887598)
```

Skewness

```
In [20]: df.skew()
```

```
Out[20]: YearsExperience    0.37956
Salary    0.35412
dtype: float64
```

```
In [21]: df['Salary'].skew()
```

```
Out[21]: np.float64(0.35411967922959153)
```

Standard Error

```
In [22]: df.sem()
```

```
Out[22]: YearsExperience    0.518125  
Salary          5005.167198  
dtype: float64
```

```
In [23]: df['Salary'].sem()
```

```
Out[23]: np.float64(5005.167198052405)
```

Z-score

```
In [24]: # for calculating Z-score we have to import a library first  
import scipy.stats as stats  
df.apply(stats.zscore) # this will give Z-score of entire dataframe
```

Out[24]:

	YearsExperience	Salary
0	-1.510053	-1.360113
1	-1.438373	-1.105527
2	-1.366693	-1.419919
3	-1.187494	-1.204957
4	-1.115814	-1.339781
5	-0.864935	-0.718307
6	-0.829096	-0.588158
7	-0.757416	-0.799817
8	-0.757416	-0.428810
9	-0.578216	-0.698013
10	-0.506537	-0.474333
11	-0.470697	-0.749769
12	-0.470697	-0.706620
13	-0.434857	-0.702020
14	-0.291498	-0.552504
15	-0.148138	-0.299217
16	-0.076458	-0.370043
17	-0.004779	0.262859
18	0.210261	0.198860
19	0.246100	0.665476
20	0.532819	0.583780
21	0.640339	0.826233
22	0.927058	0.938611
23	1.034577	1.402741
24	1.213777	1.240203
25	1.321296	1.097402
26	1.500496	1.519868
27	1.536336	1.359074
28	1.787215	1.721028
29	1.858894	1.701773


```
In [25]: stats.zscore(df['Salary'])
```

```
Out[25]: array([-1.36011263, -1.10552744, -1.419919 , -1.20495739, -1.33978143,
        -0.71830716, -0.58815781, -0.79981746, -0.42881019, -0.69801306,
        -0.47433279, -0.74976858, -0.70662043, -0.70201994, -0.55250402,
        -0.29921736, -0.37004264,  0.26285865,  0.19885989,  0.66547573,
         0.58377993,  0.82623317,  0.93861127,  1.40274136,  1.24020308,
         1.09740238,  1.51986835,  1.3590738 ,  1.72102849,  1.70177321])
```

Degree of Freedom

```
In [26]: a = df.shape[0] # this will gives us no.of rows
        b = df.shape[1] # this will give us no.of columns
        degree_of_freedom = a-b
        print(degree_of_freedom) # this will give us degree of freedom for entire dataset
```

28

Sum of squares Regression (ssr)

```
In [28]: #First we have to separate dependent and independent variables
        X=df.iloc[:, :-1].values #independent variable
        y=df.iloc[:, 1].values
        # dependent variable
        y_mean = np.mean(y) # this will calculate mean of dependent variable
        from sklearn.model_selection import train_test_split
        X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.20,random_state=0)
        from sklearn.linear_model import LinearRegression
        reg = LinearRegression()
        reg.fit(X_train,y_train)
        y_predict = reg.predict(X_test)
        SSR = np.sum((y_predict-y_mean)**2)
        print(SSR)
```

6263152884.284127

Sum of Squares Error (sse)

```
In [30]: #First we have to separate dependent and independent variables
        X=df.iloc[:, :-1].values #independent variable
        y=df.iloc[:, 1].values
        # dependent variable
        from sklearn.model_selection import train_test_split
        X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.20,random_state=0)
        from sklearn.linear_model import LinearRegression
        reg = LinearRegression()
        reg.fit(X_train,y_train)
        y_predict = reg.predict(X_test) # before doing this we have to train,test and split
        y = y[0:6]
```

```
SSE = np.sum((y-y_predict)**2)
print(SSE)
```

15274062883.9432

Sum of squares Total (sst)

```
In [32]: mean_total = np.mean(df.values)
SST = np.sum((df.values-mean_total)**2)
print(SST)
```

108429703765.82735

R-Square

```
In [33]: r_square = SSR/SST
r_square
```

Out[33]: np.float64(0.05776233510524465)

In []: